

# OligoTox-Thrombocytopenia: a curated, per-measurement dataset of oligonucleotide platelet toxicity

Narrative document — NIH/NCATS Oligonucleotide Toxicity (OligoTox) Open Data Challenge, Phase 2 (Data Generation)

**Endpoint:** thrombocytopenia / platelet toxicity · **Dataset:** 259 oligonucleotides × 1,959 graded measurements from 70 primary sources · **Licence:** CC-BY 4.0 · **Repository:** navierol/Oligos → thrombocytopenia/

## 1. Executive summary

**OligoTox-Thrombocytopenia** is an openly-releasable, per-measurement dataset of platelet-toxicity signals for therapeutic oligonucleotides. Thrombocytopenia is one of the toxicities of interest named in the Challenge announcement, and — unlike hepatotoxicity — no comparable public, structured, per-compound resource for it exists. One row is one *oligo* × *model/subject* × *delivery* × *dose* × *readout*, carrying both the **design predictors** (sequence, backbone, phosphorothioate count, sugar chemistry, conjugate) and a **graded outcome** on an ordinal 0–3 scale, together with per-row provenance and rights status.

|                       |                         |                     |                       |                                  |
|-----------------------|-------------------------|---------------------|-----------------------|----------------------------------|
| 1,959<br>MEASUREMENTS | 259<br>OLIGONUCLEOTIDES | 1,453<br>HUMAN ROWS | 70<br>PRIMARY SOURCES | 23<br>CROSS-SPECIES<br>COMPOUNDS |
|-----------------------|-------------------------|---------------------|-----------------------|----------------------------------|

### The endpoint is bimodal, and the dataset is built to preserve that

Oligonucleotide-associated thrombocytopenia is not one phenomenon. The literature consistently describes (i) a **common, mild, dose- and plasma-concentration-dependent** platelet decline that plateaus and is reversible, and (ii) a **rare, severe, idiosyncratic, immune-mediated** thrombocytopenia with treatment-emergent antiplatelet antibodies and counts below  $25 \times 10^9/L$ . A dataset that averages these together destroys the distinction a predictive model most needs. Inotersen's FDA label documents both modes in one compound — the basis of its Boxed Warning — and is the anchor against which the grading rubric was calibrated. The FDA clinical review for that product states the same typology explicitly ("Rare" vs "Common"), which is independent regulatory confirmation of the framing rather than our inference.

### Positive and negative controls

| Role                          | Exemplars                                               | Evidence                                                                                                                                                                                          |
|-------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive — severe mode        | inotersen; volanesorsen                                 | Boxed Warning; platelets $< 25 \times 10^9/L$ in 3 % with antiplatelet IgG in 3/3; one fatal intracranial haemorrhage. Volanesorsen: dose-limiting, $< 100 \times 10^9/L$ in 47 % vs 0 % placebo. |
| Positive — mild mode          | olezarsen; ISIS 104838; the (AC) <sub>n</sub> series    | Mean platelet decline of 6–10 % without bleeding; in-vitro activation with monotonic dependence on PS count.                                                                                      |
| Negative — neutral backbone   | eteplirsen, golodirsen, viltolarsen, casimersen         | PMO morpholino backbone; zero platelet content across their FDA labels; a pooled meta-analysis reports an explicit <b>zero</b> for the PMO subgroup against a 21.2 % overall rate.                |
| Negative — targeted conjugate | inclisiran, givosiran, lumasiran, vutrisiran, patisiran | Explicit reported zeros, e.g. <i>Platelets</i> $\leq 75 \times 10^9/L$ : 0 (0.0 %) in the inclisiran FDA review; an FDA-generated platelet shift table for vutrisiran/patisiran.                  |
| Negative — isosequential      | ODN2395 phosphodiester; fesomersen                      | Same base sequence as an active comparator, differing only in backbone or conjugate.                                                                                                              |
| Negative — mechanistic        | GPVI-deficient donor platelets                          | No binding or activation up to 10 $\mu M$ — a receptor-level negative that isolates the mechanism.                                                                                                |

**A discipline that shapes the negative class.** A compound whose label never mentions platelets is *not* a measured zero. Every grade-0 row is backed by a source that reports platelet monitoring having been performed and found unremarkable. Absence of mention was never converted into evidence of absence, and the code that swept the labels states this in its own output.

## 2. Main findings and conclusions

### 2.1 The dataset reproduces the field's structure–activity relationship from curation alone

Platelet effects are held to track **phosphorothioate (PS) content and protein binding** rather than the hybridisation target, mechanistically via glycoprotein VI. Without any modelling, the assembled data orders exactly as that hypothesis predicts:

| Backbone    | rows  | compounds | mean grade  | PS linkages | mean grade  |
|-------------|-------|-----------|-------------|-------------|-------------|
| PMO_neutral | 10    | 6         | <b>0.10</b> | 0           | 0.39        |
| PS_P0_mix   | 54    | 7         | 0.15        | 13–16       | 0.60        |
| full_P0     | 57    | 6         | 0.44        | 17–19       | 1.07        |
| full_PS     | 1,611 | 170       | <b>0.99</b> | 20+         | <b>1.47</b> |

Modality orders in the same direction: PMO 0.10 < GalNAc-siRNA 0.27 < siRNA 0.50 < splice-switching 0.67 < ASO gapmer 0.89 < aptamer 1.60. That a curated dataset recovers a relationship the field established experimentally is the primary evidence that it is fit to train on.

### 2.2 Five controlled comparisons — the dataset's most distinctive content

Curation deliberately sought **matched pairs and titration series**: cases where one design variable moves while the others are held fixed. These let a model learn a mechanism rather than a correlation.

| # | Comparison                                 | Variable isolated                                         | Result                                                                                      |
|---|--------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1 | ODN2395 phosphodiester vs phosphorothioate | backbone only (0 → 21 PS); identical 22-mer               | PAC-1 binding 1.65 → <b>18.94</b> MFI on human platelets                                    |
| 2 | the (AC) <sub>n</sub> series, n = 5...11   | length, hence PS count, sequence held constant            | monotonic 2.12 → 16.79 MFI; threshold between 9 and 11 linkages                             |
| 3 | ISIS 416858 vs fesomersen                  | 19 → 13 PS + GalNAc; identical sequence and 5-10-5 design | dose-dependent monkey platelet decline vs none                                              |
| 4 | volanesorsen vs olezarsen                  | GalNAc conjugate and the ~20× lower dose it permits       | dose-limiting (47 % < 100 × 10 <sup>9</sup> /L) vs a 6–10 % mean decline, no major bleeding |
| 5 | ISIS 972163 vs ISIS 972116                 | <b>one nucleoside</b> in a wing (cEt → 2'-MOE)            | 556 vs 325 K/μL in the same rat table — a 41 % difference                                   |

Comparison 4 required recovering olezarsen's sequence, which is not published as a string, by deterministic parse of the IUPAC chemical name in its FDA label. The parse returns **AGCTTCTTGTCAGCTTTAT** — identical to volanesorsen's independently published sequence, exactly as expected for the GalNAc-conjugated APOC3 analogue, with the stated molecular formula **P20S19** independently fixing full-PS chemistry at 20 residues. Two derivation paths reaching the same string is meaningful evidence against transcription artefact.

### 2.3 Human and animal evidence are separated, and 23 compounds bridge them

The announcement singles out datasets "based on in vitro human systems or able to **extrapolate data between in vitro human systems and animal data**". The dataset is **74 % human** (1,453 of 1,959 rows), and the axis is a first-class derived column, **subject\_class**, computed from study type × species and independently re-derived by the QC gate so it cannot drift.

| Human                                  | rows                        | Animal       | rows                       |
|----------------------------------------|-----------------------------|--------------|----------------------------|
| clinical (trials)                      | 1,002                       | in vivo      | 429                        |
| in vitro (platelets, PRP, whole blood) | 426                         | in vitro     | 55                         |
| ex vivo                                | 25                          | ex vivo      | 13                         |
| <b>total</b>                           | <b>1,453</b> (78 compounds) | <b>total</b> | <b>497</b> (174 compounds) |

Containing human rows and animal rows does not by itself demonstrate extrapolation. Having the *same compound* characterised on both sides does, and **23 compounds** qualify — inotersen (199 human / 22 animal rows), volanesorsen (172 / 18), mipomersen (114 / 31) among them, each with a published sequence. `data/bridge_human_animal.csv` presents them side by side with their human and animal grades. That set, not the raw totals, is what a cross-species model can be trained and validated on.

## 2.4 Mechanism must be labelled, not assumed

Imetelstat causes severe thrombocytopenia, but by **on-target telomerase-inhibitor myelosuppression**, not PS-backbone platelet binding. Left unflagged, its 93 high-grade rows made up 74 % of the zero-PS bucket and pushed that bucket's mean grade *above every phosphorothioate bucket* — inverting the headline result. It is mechanism-flagged and excluded from structure–activity analysis while remaining in the dataset. Similarly, volanesorsen's anti-*drug* antibodies carried no altered safety profile and must not be conflated with inotersen's anti-*platelet* IgG, which tracked with severe events.

### 3. How the data were produced

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This is an **in-silico curation** of pre-existing, publicly reported data — not wet-lab-generated data. "Experimental design" therefore describes source identification, extraction, harmonisation, grading, provenance and quality control. Full detail is in the accompanying methodology document.

#### 3.1 Design

- **Grain:** one row per oligo × model/subject × delivery × dose/concentration × readout. A compound measured for platelet count *and* P-selectin at one dose yields two rows.
- **Two normalised tables** joined on `oligo_id`: `oligos.csv` (identity and design predictors, 20 columns) and `measurements.csv` (outcome and context, 24 columns). A denormalised analysis view is generated, never hand-edited.
- **Coverage goal:** all modalities, all study types, the full severity range, and **well-sourced negative controls** — which are what make the chemistry hypothesis falsifiable.
- Missing values are the literal string `TBD` — never guessed, never imputed as zero.

#### 3.2 Acquisition

Sources were discovered by a **parallel multi-agent sweep across ten independent lanes** (clinical, regulatory, in-vitro human platelet, mechanism/immune, preclinical, non-ASO negatives, patents, reviews/meta-analyses, megakaryocyte models, sequence–chemistry panels), each blind to the others. Lanes that overlapped provided cross-confirmation; lanes that did not surfaced sources the others missed. Sources were ranked by **extractable per-measurement content**, not topicality, and each had to be *fetched* before being proposed. Primary sources were retrieved directly: NCBI E-utilities and Europe PMC full text plus supplementary files, the DailyMed SPL API for FDA labels, `accessdata.fda.gov` for FDA review documents, the EMA for SmPCs and EPAR assessment reports, and USPTO/Google Patents for patent full text and sequence listings.

Regulatory review documents proved the highest-value and least-mined vein: FDA multidiscipline reviews and EMA assessment reports carry per-patient platelet nadir listings, antibody cross-tabulations and dose-banded incidence grids that the corresponding journal publications do not.

#### 3.3 Computational processing

The path from extraction to published CSV is fully scripted and re-runnable (`scripts/ingest_thrombo.sh`): harvest → merge lanes in a deterministic precedence order → assemble (apply verification verdicts, deduplicate on a natural key, assign stable keys) → re-apply committed verdicts → **QC gate** → rebuild the derived view and the generated documentation sections. QC gates the round: if it fails, the derived artefacts are not regenerated, so a broken dataset cannot ship with fresh-looking documentation. The **raw curation record** — every agent's returned rows, the verification verdicts, the source sweep — is committed alongside the data, so the published tables are reproducible from their inputs rather than merely re-checkable against their citations. Rebuilding from that record reproduces both CSVs byte-for-byte.

#### 3.4 Accuracy control

- **Adversarial verification.** Extracted rows are passed to an independent agent instructed to *refute* them against the cited source. Two blocks totalling 659 rows have completed: **628 confirmed, 31 corrected, 0 rejected**. For one block every value was recomputed from the paper's raw per-replicate workbook and every mean, SD and n reproduced exactly.
- **Held-out known-answer test.** The inotersen label was parsed independently and its figures recorded *before* any extraction ran; the assembled dataset matched all six.
- **Cross-dataset sequence agreement.** Sequences were checked against a sister dataset's records, which were validated there against WHO INN nomenclature and patent sequence listings: **zero conflicts**.

**Four defects found and corrected, worth reporting because three were invisible to schema validation.** (i) A cross-source **name collision**: ODN 2395 denotes a phosphorothioate reagent everywhere except one paper that made a phosphodiester variant under the bare name — 39 rows describing an active PS compound had attached to a record asserting `ps_count = 0`, undercutting the central hypothesis. (ii) The **mechanism confound** of §2.4. (iii) A **header-misalignment** in three patent tables that silently returned red-cell counts where platelet counts were intended in 35 of 55 rows — real haematology numbers, of plausible magnitude, from the correct study. (iv) A controlled-vocabulary typo. All four produced structurally valid data; they were caught by asking whether the assembled data still reproduced a relationship the field already knows, which is why that check runs every round.

## 4. Indicators, predictors, and their distributions

### 4.1 The graded indicator

Each measurement carries an ordinal **thrombocytopenia\_grade (0–3)**, CTCAE-aligned for clinical rows with explicit in-vitro analogues so a bench readout and a trial outcome land on a comparable scale: **0** no signal at tested exposure · **1** mild/reversible,  $\geq 100 \times 10^9/L$  or activation only at supra-clinical concentration · **2** moderate,  $50\text{--}99 \times 10^9/L$  or activation at clinically relevant concentration · **3** severe,  $< 50 \times 10^9/L$ , CTCAE grade 4, antibody-mediated, bleeding attributable to thrombocytopenia, or discontinuation.

Two conventions are applied uniformly, both stated because they are the easiest places to introduce systematic error. An **incidence row is graded by the severity of the event**, not by how small the incidence is. **Bleeding is graded on attribution**: grade 3 requires haemorrhage attributable to thrombocytopenia, so a trial reporting mild bleeding at or below the placebo rate is not scored severe.

| Grade        | rows | Readout category      | rows  | Study type     | rows  |
|--------------|------|-----------------------|-------|----------------|-------|
| 0 — none     | 852  | platelet count        | 1,083 | clinical       | 1,002 |
| 1 — mild     | 499  | clinical outcome      | 226   | in vitro       | 485   |
| 2 — moderate | 388  | platelet activation   | 196   | animal in vivo | 434   |
| 3 — severe   | 220  | binding / aggregation | 236   | ex vivo        | 38    |
|              |      | immunogenicity        | 105   |                |       |
|              |      | megakaryocyte / other | 113   |                |       |

Readouts deliberately emphasise the **functional and immunological** axes over viability: 1,083 platelet-count rows, 196 activation, 236 binding/aggregation, and 105 immunogenicity rows capturing the antiplatelet-antibody mechanism that distinguishes the severe mode.

### 4.2 Predictor distribution among the 259 tested oligos

| Modality             | n   | Backbone    | n   | Conjugate   | n   | Stage          | n   |
|----------------------|-----|-------------|-----|-------------|-----|----------------|-----|
| ASO gapmer           | 190 | full PS     | 193 | none        | 228 | research panel | 135 |
| other / tool         | 45  | PS/PO mix   | 11  | GalNAc      | 15  | preclinical    | 53  |
| PMO                  | 6   | full PO     | 6   | lipid       | 5   | phase 1        | 24  |
| aptamer              | 7   | PMO neutral | 6   | PEG / other | 4   | phase 2–3      | 16  |
| splice-switching     | 5   | mixed / NA  | 4   | TBD         | 7   | approved       | 17  |
| GalNAc-siRNA / siRNA | 6   | TBD         | 39  |             |     | class review   | 8   |

**Sequences are present for 194 of 259 oligos (74 %)** and were never guessed: the remainder are proprietary, class-level aggregates, or development-code compounds with no published sequence. **67 distinct target genes** are represented. Provenance is complete — every row carries a source identifier, an exact locus (table, figure, claim or label section) and a rights class: **748 rows are public-domain** (patents, FDA/EMA), 227 are CC-BY, and 984 are reproduced as summary statistics.

## 5. The gap this addresses

Phase 1 of this Challenge was dominated by hepatotoxicity; over half the winning teams targeted liver models. Thrombocytopenia is a named toxicity of interest that **no public, structured, per-compound resource covers**, despite being the toxicity that carries a Boxed Warning on a marketed oligonucleotide, that was dose-limiting for another, and that has caused a trial death.

The gap is not that the information is secret — it is that it is **scattered and unusable**. The evidence sits in FDA multidiscipline reviews, EMA assessment reports, patent example tables and the supplementary files of individual papers, in incompatible formats, without harmonised severity grading, and without any linkage between a compound's chemistry and its platelet outcome. Nothing lets a modeller ask "what does phosphorothioate count do to platelets across every compound ever reported?"

This dataset closes that specific gap. It contributes:

- **Harmonised grading** across in-vitro, ex-vivo, animal and clinical evidence on one 0–3 scale, so sources that never shared a metric become comparable.
- **Design predictors joined to outcomes** — sequence, PS count, backbone, conjugate — which is the join no existing source performs.
- **A deliberately populated negative class** (795 grade-0 rows). Negative results are systematically under-published; sourcing them from regulatory documents, where monitoring is reported whether or not it was eventful, is a contribution in itself.
- **Controlled comparisons and a 23-compound cross-species bridge**, which are what make mechanism learnable rather than merely correlated.
- **Per-row rights tracking**, so a consumer can filter to exactly the rows they may lawfully reuse.

## 6. Using the data to build a predictive model

A worked demonstration is included (`scripts/model_demo.py`) rather than an assertion. The task: predict whether an oligo × condition shows *any* platelet effect (grade ≥ 1) from design features. Control-arm rows and the mechanism-confounded compound are excluded, leaving 1,728 rows over 228 compounds.

| Evaluation                                            | ROC-AUC      | balanced acc. | What it shows                                        |
|-------------------------------------------------------|--------------|---------------|------------------------------------------------------|
| Baseline — always predict majority                    | —            | 0.500         | reference                                            |
| <b>[A] Design features only</b> , grouped by compound | <b>0.627</b> | 0.633         | chemistry alone carries real signal                  |
| [B] Design + study context, grouped                   | 0.690        | 0.687         | +0.062 — the study-type confound, <i>not</i> biology |
| Ungrouped random row split                            | 0.752        | 0.690         | +0.081 of pure leakage                               |

Three points matter more than the score. **Grouped cross-validation is mandatory**: rows are not independent — one compound contributes nearly 200 — so a random row split lets a model memorise compounds rather than learn chemistry, and we quantify that illusion at +0.081 ROC-AUC. **Chemistry alone beats baseline**, with `ps_count` the single most important feature, so the dataset supports its own stated hypothesis under a split that forbids memorisation. And the **A-versus-B gap is reported precisely so it is not banked as chemistry performance** — it reflects severe events being observed in trials and not in dishes.

Beyond this baseline the dataset supports: **sequence-level models** (194 sequences with graded outcomes), **cross-species translation** on the 23-compound bridge, **dose–response modelling** from the banded clinical incidence grids, and — the clinically decisive task — **separating the mild dose-dependent mode from the severe immune-mediated mode**, for which the `immunogenicity` readout category provides the labels.

**What we do not claim.** This is a feasibility demonstration, not a validated clinical predictor. Grade remains partly confounded with study type; the severe idiosyncratic mode is antibody-driven and is *not* expected to be predictable from design at all; and 32 % of rows have completed adversarial verification, so the remainder should be treated as curated-but-unverified. These limits are documented in the methodology rather than left for a reader to discover.

